Tableau Rules Cheat Sheet

Phil 305 — Propositional and Modal Logic

Part 1: Propositional Connectives

Each connective has a rule for when the sentence is **true** ($\mathbb{T}$) and when it is **false** ($\mathbb{F}$). Non-branching rules add lines below; branching rules split into two branches (shown side by side).

Negation ($\neg$)

$$\begin{array}{c} \mathbb{T}\,\neg A \\ \Downarrow \\ \mathbb{F}\,A \end{array} \qquad \begin{array}{c} \mathbb{F}\,\neg A \\ \Downarrow \\ \mathbb{T}\,A \end{array}$$

Both rules are non-branching.

Conjunction ($\wedge$)

$$\begin{array}{c} \mathbb{T}\,A \wedge B \\ \Downarrow \\ \mathbb{T}\,A \\ \mathbb{T}\,B \end{array} \qquad \begin{array}{c} \mathbb{F}\,A \wedge B \\ \swarrow \quad \searrow \\ \mathbb{F}\,A \quad | \quad \mathbb{F}\,B \end{array}$$

True: non-branching (both conjuncts true).
False: branching (at least one conjunct false).

Disjunction ($\vee$)

$$\begin{array}{c} \mathbb{T}\,A \vee B \\ \swarrow \quad \searrow \\ \mathbb{T}\,A \quad | \quad \mathbb{T}\,B \end{array} \qquad \begin{array}{c} \mathbb{F}\,A \vee B \\ \Downarrow \\ \mathbb{F}\,A \\ \mathbb{F}\,B \end{array}$$

True: branching (at least one disjunct true).
False: non-branching (both disjuncts false).

Conditional ($\rightarrow$)

$$\begin{array}{c} \mathbb{T}\,A \rightarrow B \\ \swarrow \quad \searrow \\ \mathbb{F}\,A \quad | \quad \mathbb{T}\,B \end{array} \qquad \begin{array}{c} \mathbb{F}\,A \rightarrow B \\ \Downarrow \\ \mathbb{T}\,A \\ \mathbb{F}\,B \end{array}$$

True: branching (antecedent false or consequent true).
False: non-branching (antecedent true, consequent false).

Closure: A branch closes when it contains both $\mathbb{T}\,A$ and $\mathbb{F}\,A$ for some sentence A. A tableau closes when *every* branch closes.

Strategy: Apply non-branching rules before branching rules.

Part 2: Modal Rules in K

In **K**, worlds are numbered with dot notation: x, $x.y$, $x.y.z$, etc. World x can access world $x.y$ (and only those forced by the dot structure).

True $\Box$ (open-ended, reusable)

If $\mathbb{T}\,x\colon \Box A$ appears on the branch, then for *every* world $x.y$ on the branch, add:

$$\mathbb{T}\,x.y\colon A$$

Apply this each time a new $x.y$ appears.

False $\Diamond$ (open-ended, reusable)

If $\mathbb{F}\,x\colon \Diamond A$ appears on the branch, then for *every* world $x.y$ on the branch, add:

$$\mathbb{F}\,x.y\colon A$$

Apply this each time a new $x.y$ appears.

True $\Diamond$ (one-shot)

If $\mathbb{T}\,x\colon \Diamond A$ appears on the branch, introduce a *new* world $x.y$ (not yet on the branch) and add:

$$\mathbb{T}\,x.y\colon A$$

False $\Box$ (one-shot)

If $\mathbb{F}\,x\colon \Box A$ appears on the branch, introduce a *new* world $x.y$ (not yet on the branch) and add:

$$\mathbb{F}\,x.y\colon A$$

Key distinction: True $\Box$ / False $\Diamond$ are *reusable*—they apply to every accessible world that appears. True $\Diamond$ / False $\Box$ are *one-shot*—they introduce a single new world.

Tableau Rules Cheat Sheet (cont.)

Additional Rules for T, 4, and B

Part 3: Rules Beyond K

These rules are **added to** the K rules. They only affect the open-ended rules (True $\Box$ and False $\Diamond$). The one-shot rules (True $\Diamond$ and False $\Box$) stay the same in all logics.

T Rules (Reflexivity: every world accesses itself)

T-Box Rule: If $\mathbb{T}\, x\colon \Box A$ is on the branch, add:

$$\mathbb{T}\, x\colon A$$

(Since x accesses itself, A must hold at x too.)

T-Diamond Rule: If $\mathbb{F}\, x\colon \Diamond A$ is on the branch, add:

$$\mathbb{F}\, x\colon A$$

(Since x accesses itself, A must fail at x too.)

4 Rules (Transitivity: if x accesses $x.y$, then x accesses everything $x.y$ accesses)

4-Box Rule: If $\mathbb{T}\, x\colon \Box A$ is on the branch and $x.y$ is on the branch, add:

$$\mathbb{T}\, x.y\colon \Box A$$

(The whole $\Box A$, not just A, carries forward.)

4-Diamond Rule: If $\mathbb{F}\, x\colon \Diamond A$ is on the branch and $x.y$ is on the branch, add:

$$\mathbb{F}\, x.y\colon \Diamond A$$

(The whole $\Diamond A$ false, not just A false, carries forward.)

B Rules (Symmetry: if x accesses $x.y$, then $x.y$ also accesses x)

B-Box Rule: If $\mathbb{T}\, x.y\colon \Box A$ is on the branch, add:

$$\mathbb{T}\, x\colon A$$

(Since $x.y$ can look back to x, A must hold at x.)

B-Diamond Rule: If $\mathbb{F}\, x.y\colon \Diamond A$ is on the branch, add:

$$\mathbb{F}\, x\colon A$$

(Since $x.y$ can look back to x, A must fail at x.)

Part 4: Quick Reference — Combining Logics

Logic	Rules Used	Frame Condition
K	K rules only	No restrictions on R
KT	K + T rules	Reflexive (xRx)
K4	K + 4 rules	Transitive (if xRy and yRz then xRz)
KB	K + B rules	Symmetric (if xRy then yRx)
S4 (= KT4)	K + T + 4 rules	Reflexive + Transitive
KTB	K + T + B rules	Reflexive + Symmetric
S5 (= KT4B)	K + T + 4 + B rules	Equivalence relation (= universal)

Characteristic axioms:

- **T:** $\Box A \to A$ (what's necessary is actual)
- **B:** $A \to \Box\Diamond A$ (what's actual is necessarily possible)
- **4:** $\Box A \to \Box\Box A$ (what's necessary is necessarily necessary)

Reminder for S5 (simplified tableaux): In S5, every world accesses every world. So:

- $\mathbb{T}\,\Box A$ at any world $\Rightarrow$ $\mathbb{T}\, A$ at *every* world.
- $\mathbb{F}\,\Diamond A$ at any world $\Rightarrow$ $\mathbb{F}\, A$ at *every* world.
- $\mathbb{T}\,\Diamond A$ at any world $\Rightarrow$ introduce one new world with $\mathbb{T}\, A$ (can't reuse a world where A is already false).
- $\mathbb{F}\,\Box A$ at any world $\Rightarrow$ introduce one new world with $\mathbb{F}\, A$ (can't reuse a world where A is already true).